Page 1: Introduction to Plant Structures

Plants are living organisms that cover much of the Earth. Unlike animals, plants cannot move from place to place.

Instead, they have specialized structures that help them survive in their environment.

Roots anchor the plant firmly into the soil and absorb water and essential minerals.

The stem supports the leaves and flowers, carrying water and nutrients upward from the roots.

Leaves are the food factories of the plant, collecting sunlight to produce energy.

Flowers help the plant reproduce by creating seeds for the next generation of plants.

Page 2: What Plants Need to Grow

All green plants require four basic things to survive: sunlight, water, air, and soil nutrients.

Sunlight provides the thermal energy needed for chemical reactions inside the green leaves.

Water carries dissolved nutrients through the stem up to the leaves and flowers.

From the air, plants absorb a gas called carbon dioxide through tiny pores on their leaves.

Healthy soil provides essential minerals like nitrogen, phosphorus, and potassium.

If any of these four elements is missing, a plant cannot grow properly.

Page 3: Photosynthesis - How Plants Make Food

Photosynthesis is the remarkable process where green plants make their own food.

Inside plant leaf cells are microscopic green structures called chloroplasts.

Chloroplasts contain a bright green pigment known as chlorophyll, which traps sunlight.

The plant combines water from the soil with carbon dioxide from the air using sunlight energy.

This chemical reaction produces glucose, a natural sugar that feeds the plant, and releases oxygen gas into the air.

Humans and animals breathe the oxygen that plants release into the atmosphere.

Page 4: Plant Parts We Eat Every Day

Many delicious and healthy foods that people eat are actually different parts of plants.

Carrots, radishes, and sweet potatoes are underground roots swollen with stored nutrients.

Celery stalks and asparagus spears are edible stems that hold plants upright.

Spinach, kale, and lettuce are edible green leaves packed with vitamins.

Broccoli heads and cauliflower are clusters of tightly closed flower buds.

Apples, oranges, and tomatoes are fruits that protect the plant's seeds.

Page 5: Flower Pollination and Seeds

Flowers are brightly colored and scented to attract pollinating insects like bees and butterflies.

When a honeybee lands on a flower to drink sweet nectar, yellow pollen grains stick to its fuzzy body.

The bee flies to another flower, transferring the pollen so that seeds can begin developing.

Once fertilized, the flower petals fall away, and a fruit or seed pod forms around the seeds.

Seeds can travel by wind, floating on water, or catching on animal fur to find new soil.

Page 6: Math in Nature - Flower Petal Multiplication

Nature is full of mathematical patterns and repeated arrays.

Imagine you are walking through a botanical garden and spot 3 bright yellow sunflowers.

If each sunflower has 5 large petals, how many petals are there in total?

We can calculate this using multiplication: 3 * 5 = 15 petals.

If you discover 4 clover stems and each stem has 3 green leaves, you can multiply: 4 * 3 = 12 leaves.

Multiplication is a fast way to add equal groups together in nature.

Page 7: Tree Rings and Calculating Tree Age

Trees grow larger in diameter every year by adding a new layer of wood under their bark.

When a tree trunk is cut across, these yearly growth layers appear as concentric circles called tree rings.

A light ring grows during the warm spring, and a dark ring forms during late summer and autumn.

Together, one light ring and one dark ring equal one complete year of growth.

If an oak tree stump has 10 inner rings and 4 outer rings, its total age is: 10 + 4 = 14 years old.

Counting rings tells scientists about past climate, rainfall, and drought.

Page 8: Desert and Rainforest Plant Adaptations

Different plants have adapted unique features to survive in extreme environments across the planet.

In dry deserts, the saguaro cactus has thick, waxy green stems that store gallons of water.

Instead of broad leaves that lose moisture quickly, cacti have sharp spines for defense and shade.

In tropical rainforests, plants receive abundant rain but compete fiercely for sunlight under the dense canopy.

Many rainforest vines climb high into the tallest trees to reach the bright sun.

Page 9: Key Chapter Vocabulary Review

1. Chlorophyll: The green pigment in leaves that absorbs sunlight for photosynthesis.

2. Glucose: The simple sugar that plants produce to nourish their cells.

3. Pollination: The transfer of pollen between flowers by pollinators like bees or wind.

4. Roots: The underground plant organs that absorb water and anchor the plant in soil.

5. Oxygen: The vital gas produced during photosynthesis that animals breathe to survive.

Page 10: Science Investigation and Math Practice

Let us test our math and science skills with realistic field observations:

Question Example 1: A farmer picks 6 pea pods from the garden vine. Each pea pod contains 4 round green peas.

To find the total number of peas, we compute: 6 * 4 = 24 peas.

Question Example 2: A seedling had 8 leaves on Monday. Over the week, it sprouted 7 new leaves.

The total number of leaves is now: 8 + 7 = 15 leaves.

Remember: Always observe carefully, count equal groups, and take good care of growing plants!